

Ejercicio en clase

• $W = \text{Morelia}$, $X = \text{More}$, $Y = \text{lia}$

Encontrar: $W^* = (XY)^* = \{a, i, l, e, r, o, M\}^* \rightarrow \{ \text{Morelia} \}$

$W = (w^*)^* = \{ \text{Morelia} \}^*$

(Morel)ai
(More)ail
(Mor)aile
(Mo)ailer
(M)ailero

$w^* = \{ \text{ailero} \} M$

$w' = \text{ailero} M$

• $L = \{a, b, c\}$ y $M = \{1, 2\}$

Encontrar:

$LM = \{(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)\}$

$L \{ \epsilon \} = \emptyset$

• $L = \{011\}$ sobre $\Sigma = \{0, 1\}$

$L^0 = \{ \epsilon \}$

$L^3 = \{011, 011, 011\}$

$L^1 = \{011\}$

$L^4 = \{011, 011, 011, 011\}$

$L^2 = \{011, 011\}$

• $L = \{ \text{Guatemala, Quetzaltenango} \}$; $\Sigma = \{a, b, c, \dots, z\}$ encontrar L^*

$W^* \rightarrow (\text{Guatemala, quetzaltenango})^* \rightarrow (\text{Guatemala, quetzalten})^* \text{ognq}$

$\rightarrow (" , quetzaltenan)^* \text{og}$ } $\rightarrow (" , quetzalte)^* \text{ognan}$

$\rightarrow (" , quetzaltena)^* \text{ogn}$ } $\rightarrow (" , quetzal)^* \text{ognant}$

• $\Sigma = \{a, b\}$, $L = \{a^*bb^*\}$

$L^* = L^0 \cup L^1 \cup L^2 \cup \dots = \{ \lambda, abb, (abb, abb), \dots \}$

$L^+ = L^1 \cup L^2 \cup L^3 \cup \dots = \{ abb, (abb, abb), (abb, abb, abb), \dots \}$

• $\Sigma = \{0, 1\}$ y $L = \{011, 011011, 011011011, \dots\}$
 Encontrar $M = \{a^n / a = 011; n \in \mathbb{N}\}$
 $M = \{a^1, a^2, a^3, \dots\} = \{011, 011011, 011011011, \dots\}$

Demstrar $L \subset M$
 La potencia de a da como resultado el lenguaje

$L \cup M = \{011, 011011, 011011011, \dots\}$
 $L \cap M = \{011, 011011, 011011011, \dots\}$
 $L - M = \{011, 011011, 011011011, \dots\}$
 $L^c = \{0, 1\}$

• Sean: $\Sigma = \{a, b, c\}$
 $L = \{\lambda, abc, aab, aabb, bbc, cb, cab\}$; $M = \{aa, ab, aabb, bab, cb, cc\}$

$L \cup M = \{\lambda, abc, aab, aabb, bbc, cb, cab, aa, ab, bab, cc\}$
 $L \cap M = \{aabb, cb\}$
 $L - M = \{\lambda, abc, aab, bbc, cab\}$
 $L^c = \{a, b, c\}$

→ (Guatemala, Quetz) o gnant la } → (Guatemala) o gnant la z te u a,
 → (" , Quetz) o gnant la } → (Guatemala) o gnant la z te u a, a
 → (" , Quetz) o gnant la z } → (Guatemala) o gnant la z te u a, a l
 → (" , Que) o gnant la z t } → (Guatemala) o gnant la z te u a, a l a
 → (" , Qu) o gnant la z te } → (Guatemala) o gnant la z te u a, a l a m
 → (" , Q) o gnant la z te u } → (Guatemala) o gnant la z te u a, a l a m e
 → (" ,) o gnant la z te u } → (Guatemala) o gnant la z te u a, a l a m e t

- | | | | |
|--|----|----|----|
| | DD | MM | AA |
|--|----|----|----|
- (Ga)ognant lazteuQ, alameta
 - (G)ognant lazteuQ, alametau
 - (λ)ognant lazteuQ, alameta ∪ G

$$L^{\pm} = \{ \text{ognant lazteuQ, alameta} \cup G \}$$